

Acute Kidney Injury Masked by Malnutrition: A Case Report and the Problem of Protein

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Abstract

Acute kidney injury (AKI) is common among critically ill patients. There are a number of nutrition considerations in the management of AKI, including fluid balance, electrolyte and acid-base disturbances, protein provision, and management of comorbid conditions. The optimal amount of protein provision for patients with AKI who are not on renal replacement therapy (RRT) has been a topic of debate for years. Excessive protein provision may contribute to azotemia in these patients, but inadequate protein intake may harm nutrition status and result in poorer clinical outcomes. This review discusses a patient case of AKI masked by malnutrition and muscle loss and reviews the current literature on optimal protein intake in AKI (not on RRT). Based on a structured search strategy, 4 articles were reviewed. We conclude that the available evidence suggests that significant restrictions in protein intake are not necessary for those critically ill patients with AKI. However, the studies reviewed here showed significant heterogeneity in protein dose and delivery, estimation of protein needs, patient population, and definition of AKI, and thus further research is needed to systematically determine the optimal dose of protein for critically ill adults with AKI. (*Nutr Clin Pract*. 2019;34:735–750)

Keywords

acute kidney injury; critical illness; enteral nutrition; malnutrition; protein

Introduction

Acute kidney injury (AKI) occurs in approximately 14% of adult hospital admissions.¹ AKI is especially common in critically ill patients, affecting approximately 35% of those admitted to the intensive care unit (ICU).² AKI is associated with significant morbidity and mortality among hospitalized patients, and management is generally aimed at reducing renal injury and preserving remaining kidney function.³ The kidney damage incurred with AKI is potentially reversible; however, the relationship between AKI and long-term prognosis is complex, and, because of a number of factors, clinical outcomes vary widely.^{4,5}

Pathophysiology

AKI is characterized by an abrupt decline in glomerular filtration rate (GFR), with a subsequent accumulation of nitrogenous waste products (including serum urea nitrogen and creatinine).⁶ AKI is classified based upon etiology as either prerenal, renal (intrinsic), or postrenal (obstructive).^{6,7} Prerenal conditions such as hypovolemia, impaired cardiac output, reduced vascular resistance or peripheral vasodilation, and renal vasoconstriction may reduce renal perfusion and contribute to the development

of AKI.⁶ These conditions are detailed in Table 1. Prerenal injuries may subsequently lead to intrinsic renal injuries if not quickly corrected.⁷ Renal (or intrinsic) injuries describe structural damage within the kidney; these are classified as either tubular, glomerular, interstitial, or vascular damage (see Table 1).⁶ Postrenal etiologies include obstructions of the urinary tract, which increases intratubular pressure and reduce GFR.⁶ Obstructions can reduce renal blood flow and stimulate inflammatory processes, which can further impair kidney function.⁶

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Table 1. Conditions That May Contribute to the Development of AKI.^{6,7,10}

Prerenal	Hypovolemia	Vomiting, diarrhea, reduced oral intake, hemorrhage, third spacing, burns, excessive sweating, diuresis, pancreatitis
	Impaired cardiac output	CHF, cardiac tamponade, severe pulmonary hypertension, pulmonary embolus, myocardial infarction, abdominal compartment syndrome
	Reduced vascular resistance	Peripheral vasodilation from sepsis, vasodilating medications, autonomic neuropathy, anaphylaxis, drug overdose
	Renal vasoconstriction	Vasoconstricting medications, hypercalcemia, hepatorenal syndrome, abdominal compartment syndrome
Renal (intrinsic)	Tubular damage (ATN)	Ischemic (significant reduction in renal perfusion), nephrotoxic (toxicity from aminoglycosides, amphotericin B, cis-platinum, radiocontrast media, hemoglobin from hemolysis, myoglobin from rhabdomyolysis)
	Glomerular damage	Severe acute glomerulonephritis (idiopathic, or secondary to SLE, bacterial endocarditis, or Wegener's granulomatosis); HIV-associated nephropathy
	Interstitial damage	Acute interstitial nephritis (allergic reactions, or infections such as leptospirosis, legionella, Hanta virus, or rarely pyelonephritis), rhabdomyolysis, massive hemolysis, tumor lysis
	Vascular damage	Malignant hypertension, preeclampsia/eclampsia, HUS/TTP, atheroembolic disease, renal artery stenosis, scleroderma, cholesterol emboli
Postrenal (obstructive)	Urinary tract obstructions	BPH, prostate cancer, gynecologic cancers (especially cervical cancer), retroperitoneal fibrosis, ureteral stones, neurogenic bladder, papillary necrosis, medication precipitation causing intratubular obstruction

AKI, acute kidney injury; ATN, acute tubular necrosis; BPH, benign prostatic hypertrophy; CHF, congestive heart failure; HUS, hemolytic-uremic syndrome; SLE, systemic lupus erythematosus; TTP, thrombotic thrombocytopenia.

The reduction in renal blood flow that occurs as a result of the etiologies in Table 1 eventually causes ischemia and renal cell death.⁷ This ischemic injury produces a number of adverse events, including cytokine, free radical, and enzyme production; endothelial activation and leukocyte adhesion; and the stimulation of coagulation processes and apoptosis.⁷ Even after adequate renal blood flow is restored, these events may continue to cause cellular damage.⁷ GFR is further reduced by intratubular obstructions caused by the sloughing of dying cells, as well as the backflow of glomerular filtrate that occurs from impaired cellular tight junctions between damaged renal tubular cells.⁷

Acute tubular necrosis (ATN) is the most common cause of intrinsic AKI, resulting from prerenal AKI.⁸ ATN generally occurs in 3 phases: the initiation phase, the maintenance phase, and the recovery phase.⁸ The initiation phase occurs during the ischemic insult in which renal blood flow is significantly reduced, tubular cells are damaged, and the cascade of adverse events described above is triggered.^{7,8} Ischemia also reduces the production of vasodilating chemicals by tubular epithelial cells, thereby worsening vasoconstriction and hypoperfusion.⁸ During the maintenance phase (typically lasting 1–2 weeks), GFR stabilizes at a reduced level, and various complications may occur.⁸ These complications may include acid-base and electrolyte abnormalities, uremia, volume overload, and, less commonly, anemia, in-

fections, respiratory dysfunction, cardiac dysfunction, and chronic kidney disease (CKD). During the recovery phase, tubular epithelial cells regenerate, sometimes resulting in a large urine volume and significant diuresis.⁸

Diagnosis

Prompt diagnosis of AKI is crucial to help facilitate early intervention, limit nephrotoxic factors, and ultimately help to reverse kidney damage.⁹ AKI etiology is also important to evaluate to help guide medical intervention.¹⁰ The cause(s) of AKI is (are) typically determined using a combination of blood tests, urinalysis, and renal ultrasound.⁸ The fractional excretion of sodium and urea in urine can be useful in identifying prerenal AKI.¹⁰ Renal tubular epithelial cells and granular casts may be seen via urinary microscopy in cases of ATN.¹⁰ Urine may also be evaluated for proteinuria, hematuria, or urine sediment, all of which help to identify the underlying cause of AKI or warrant further investigation with kidney biopsy.¹⁰ Renal ultrasound is used to rule out obstructive or postrenal causes of AKI.¹⁰

Patient-reported symptoms can also help determine whether the kidney injury is acute or chronic, as well as aiding in establishing the etiology of AKI.⁷ Physical examination may reveal any of the manifestations of AKI and/or AKI etiologies listed in Table 2.⁷ Urine output may or may

Table 2. Manifestations of Acute Kidney Injury (AKI) or Possible AKI Etiologies Revealed in Physical Examination.^{7,52}

System	Signs	May Possibly Represent:
Skin	Livido reticularis, digital ischemia, palpable purpura, or butterfly rash Maculopapular rash Track marks	Systemic vasculitis Allergic interstitial nephritis Intravenous drug use, which may be associated with endocarditis
Eyes and ears	Eyes: Keratitis, iritis, uveitis, or dry conjunctivae Eyes: Jaundice Eyes: Band keratopathy Eyes: Microaneurysms, retinal edema, hard exudates, cotton wools spots, venous loops, venous bleeding; focal intraretinal periarteriolar transudates, arteriosclerosis, arteriovenous nicking, retinal hemorrhages Ears: Hearing loss Ears: Mucosal or cartilaginous ulcerations	Autoimmune vasculitis Liver disease Hypercalcemia, multiple myeloma Diabetic retinopathy, hypertensive retinopathy Alport disease or aminoglycoside toxicity Wegener granulomatosis
Cardiovascular	Atrial fibrillation Murmurs Pericardial friction rub Increased jugulovenous distention, rales, S ₃	Thromboemboli Endocarditis Uremic pericarditis Heart failure
Abdomen	Pulsatile mass, bruit Abdominal or costovertebral angle tenderness Pelvic or rectal masses, enlarged prostate, bladder distention	Atheroemboli Nephrolithiasis, papillary necrosis, renal artery/vein thrombosis Urinary obstruction
Pulmonary	Limb ischemia, edema Rales Hemoptysis	Rhabdomyolysis Goodpasture syndrome, Wegener granulomatosis Wegener granulomatosis

Table 3. The Definition and Classification of AKI, According to KDIGO Guidelines.^{9,11}

AKI Stage	Serum Creatinine	Urine Output
Stage I	Increased by ≥ 0.3 mg/dL or 1.5–1.9 times from baseline	<0.5 mL/kg/h for 6–12 hr
Stage II	Increased to 2.0–2.9 times from baseline	<0.5 mL/kg/h for ≥ 12 hr
Stage III	- Increased to ≥ 3.0 times from baseline; or - Serum creatinine ≥ 4.0 mg/dL; or - Requiring RRT; or - GFR <35 mL/min per 1.73 m² (in patients <18 years old)	<0.3 mL/kg/hr ≥ 24 hr; or - Anuria ≥ 12 hr

AKI, acute kidney injury; GFR, glomerular filtration rate; hr, hour; KDIGO, Kidney Disease Improving Global Outcomes; RRT, renal replacement therapy.

not be affected by AKI; however, assessing urine output aids in the differential diagnosis of AKI.⁷ Urine output of <100 mL/d (anuria) suggests possible urinary obstruction, renal artery obstruction, progressive glomerulonephritis, or bilateral diffuse renal cortical necrosis.⁷ Urine output between 100 mL and 400 mL/d (oliguria) may be associated with hepatorenal syndrome or prerenal failure.⁷ Urine output of >400 mL/d (nonoliguria) is suggestive of acute interstitial nephritis or glomerulonephritis, ATN, partial obstructive nephropathy, contrast-induced nephropathy, or rhabdomyolysis.⁷ According to Kidney Disease Improving

Global Outcomes (KDIGO), AKI may be defined by any of the following criteria:^{9,11}

- Serum creatinine increased by at least 0.3 mg/dL within 48 hours,
- Serum creatinine increased by at least 1.5 times from baseline within the past 7 days, or
- Urine output <0.5 mL/kg/h for at least 6 hours.^{9,11}

The KDIGO classifications of AKI shown in Table 3 are generally well accepted, but there are some limitations to

this system of classification.^{9,11} There are some conditions in which the serum creatinine and urine output criteria are not met, and yet clear kidney injury may still be present (eg, in muscle wasting or liver disease, in which creatinine generation is reduced), or when the criteria may be met without the presence of AKI (eg, the progressive rise in creatinine associated with CKD, or oliguria associated with the acute temporary release of antidiuretic hormone postoperatively).⁹ Patients may also lack a known baseline creatinine value, making creatinine-based diagnosis challenging.⁹ Additionally, using weight-based criteria for urine output may be misleading in patients with obesity, and it has been suggested that an ideal body weight (IBW) should be utilized in these patients to gauge urine output.^{9,12} Other biomarkers (including cystatin C, tissue injury metallopeptidase 2, insulin-like growth factor binding protein 7, neutrophil gelatinase-associated lipocalin, kidney injury molecule 1, interleukin 18, and liver-type fatty acid binding protein) have been proposed to improve the identification of AKI, though their clinical utility in practice is still unclear.¹⁰ Cystatin C appears to be among the more promising of these new biomarkers; a recent meta-analysis reported that cystatin C demonstrated high sensitivity and specificity in identifying AKI.¹³

Management

Treatment approaches for AKI are often dependent upon the underlying etiology.¹⁰ Thus, the first step in managing AKI should be identifying and treating the underlying cause. Medications must then be reviewed for potential nephrotoxic effects and adjusted if necessary.¹⁰ Table 4 lists potentially nephrotoxic medications.^{10,14} Generally, nonsteroidal anti-inflammatory medications and angiotensin-converting enzyme inhibitors/angiotension receptor blockers should be discontinued in cases of AKI.¹⁰ Intravenous iodinated contrast must be used cautiously in patients with AKI, with the need for contrast imaging weighed against its potential nephrotoxic effects.¹⁰

Maintaining hemodynamic stability in patients with AKI is crucial to help maintain renal perfusion and prevent fluid overload.¹⁰ Initial fluid resuscitation is often required in critically ill patients, and this early volume expansion is considered theoretically beneficial for those with AKI because of the improvement of cardiac output conferred by expanded intravascular volume.¹⁰ After initial fluid resuscitation, intravenous fluids should be used carefully in patients with AKI to prevent fluid overload.¹⁰ The choice of intravenous fluid in AKI is a controversial one. Colloid solutions, such as albumin, hydroxyethyl starches, and gelatins, have been studied for use in the fluid resuscitation of critically ill patients. Though intravenous albumin has been shown to reduce the risk of AKI in critically ill adults with end-stage liver disease, the data on other

Table 4. Potentially Nephrotoxic Medications.^{10,14}

Category	Medication
Analgesic	Acetaminophen Aspirin Cocaine Colchicine Methadone NSAID Phenacetin
Antiarrhythmic	Digoxin Gold Na thiomalate
Anticoagulants	Heparin Novel anticoagulants
Anticonvulsant	Phenytoin
Antidepressant	Amitriptyline
Antifungal	Amphotericin B
Antihistamine	Diphenhydramine
Antihypertensive	ACEI, ARB
Antimicrobial	Aminoglycosides B lactams Carbenicillin Cephalosporin Penicillin Pentamidine Puromycin Quinolones Rifampin Sulfonamides Tetracycline Vancomycin
Antineoplastic	Cisplatin Cyclophosphamide Interferon-alfa Methotrexate Mitomycin-C
Antiplatelet	Clopidogrel
Antipsychotic	Haloperidol Lithium
Antirheumatic	D-penicillamine
Antiviral	Acyclovir Ganciclovir Indinavir
Bisphosphonate	Pamidronate acid
Corticosteroid	Cortisone
Diuretic	Acetazolamide Furosemide -azides
Gastrointestinal	Cimetidine Lansoprazole Omeprazole Pantoprazole Ranitidine
Hypoglycemic	Cholpropamide Phenformin Tolbutamide
Hypouricemic	Allopurinol
Immunosuppressive	Cyclosporine Tacrolimus

(continued)

Table 4. (continued)

Category	Medication
Fat lowering	Statins
Muscle relaxant	Baclofen Quinine
Psychostimulant	Methamphetamine
Sedative	Benzodiazepines

ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; NSAID, nonsteroidal anti-inflammatory drug.

colloids have been mixed, and their use is not routinely recommended.¹⁰ The use of chloride-rich solutions (such as 0.9% NaCl) may increase AKI risk, as these solutions increase the risk of hyperchloremic metabolic acidosis and thus may increase renal vascular resistance, increase renin activity, and reduce GFR.¹⁰ Balanced salt solutions (which may include a mix of chloride, acetate, or other components; for example, Lactated Ringers or Plasma-Lyte A solutions) may reduce the risk of AKI and adverse kidney events.¹⁰

There are currently no consistently recommended pharmacotherapies for the prevention or management of AKI. KDIGO currently recommends against the use of diuretics, except in the case of fluid overload, as data have not consistently shown a benefit to diuretic use in AKI.¹⁰ Other medications have been proposed or are currently being studied, but because of the heterogeneous nature of AKI, no pharmacotherapies have been shown to be beneficial in its broad management.¹⁰ Associated conditions that occur as a result of AKI, including hyperkalemia, metabolic acidosis, fluid overload, or symptoms of uremia, should be medically managed to prevent further kidney injury.¹⁰

In some cases of AKI, renal replacement therapy (RRT) may be required. The timing of RRT initiation is still under some debate, with some studies showing benefit to earlier RRT initiation and others showing no clear benefit to earlier therapy.¹⁰ Continuous RRT and intermittent hemodialysis (IHD) have been shown to be equally effective in the broad population of patients with AKI. The treatment decision for which RRT modality to use should be based upon the individual patient's hemodynamic stability, fluid balance, and bleeding risk, as well as the facility's resources.¹⁰ However, even with the use of IHD, the duration of dialysis treatment may need to be extended to ensure dialysis adequacy, because of the catabolic nature of AKI in critical illness.¹⁰ Dialysis adequacy is assessed using the urea reduction ratio (Kt/V) and should be routinely monitored.¹⁰ Once initiated, RRT is then continued until (1) kidney function has improved enough to meet demands, (2) the underlying condition that triggered the AKI has resolved, or (3) continued RRT is no longer consistent with the patient/family's goals of care.¹⁰

Nutrition Implications

AKI in critically ill patients presents a number of nutrition concerns. As a catabolic and proinflammatory condition, AKI affects metabolism and can significantly alter nutrition status. Nutrition status can also affect clinical outcomes in AKI; observational studies have shown that malnutrition is associated with increased morbidity and mortality among patients with AKI and that deficits in nutrition intakes can negatively affect renal recovery.¹⁵⁻¹⁷ Because the kidneys normally maintain acid-base, electrolyte, and fluid balance; perform gluconeogenesis; secrete erythropoietin; and activate vitamin D, the impairment of renal function requires adjustment in the provision of various macronutrients and micronutrients.¹⁸ Additionally, the need for adequate protein administration in the setting of a catabolic condition must be weighed against the risk of azotemia in patients with AKI who have not yet started RRT. As the understanding of the relationship between nutrition support and AKI evolves, the literature must be revisited to ensure that nutrition support provision in clinical practice is evidence-based and best poised to improve patient outcomes.

Patient Case

An 88-year-old male was admitted to the hospital; he had a history of cardiac arrest 5 months earlier. Following the cardiac arrest, he was resuscitated and spent 3 weeks in the ICU at an outside hospital; he required tracheostomy and percutaneous endoscopic gastrostomy placement and was eventually discharged to a long-term acute-care (LTAC) facility for ventilator weaning. The patient was weaned from the ventilator after 2 months in the LTAC facility and was discharged to an extended care facility (ECF) on tracheostomy collar. Prior to his cardiac arrest, the patient was living with his daughter and was independent with activities of daily living. He had no reported history of substance abuse. The patient's daughter is a physician, as is his son-in-law. Since his cardiac arrest, the patient was nonverbal and minimally communicative. The patient's daughters indicated that he communicated by blinking. The patient's other past medical/surgical history was significant for congestive heart failure, bilateral pleural effusions, idiopathic thrombocytopenic purpura, type 2 diabetes (reportedly newly diagnosed since cardiac arrest and subsequent hospitalization), CKD, hypothyroidism, mitral valve replacement, and pacemaker placement.

The patient reportedly had an episode of vomiting and suspected aspiration while at the ECF and was started on vancomycin, metronidazole, and levofloxacin. After several days of antibiotic therapy, his vancomycin level was significantly elevated at >60 mg/L. He developed elevated serum urea nitrogen and creatinine levels, as well as worsening respiratory function, and was admitted to the hospital for

evaluation. The patient was seen jointly by the nephrologist and the registered dietitian on the evening of admission.

Upon initial assessment, the patient was found to be minimally responsive. The patient's daughters were at the bedside and provided much of the patient's history. He had been receiving enteral feedings of a standard formula with fiber at the outside hospital during his ICU stay, though duration and amount received were unknown. The patient's family reported that he received a standard formula with fiber until his blood sugar became elevated, at which point the enteral feeding was changed to a carbohydrate-controlled formula (designed for people with diabetes) with a goal of 85 mL/h (though the actual amount provided was unknown). The patient's daughters reported that his usual weight prior to his cardiac arrest was consistent with the bed-scale weight recorded at the time of the current hospital admission. Upon nutrition-focused physical exam, the patient was found to have mild to moderate subcutaneous fat loss in the orbital, neck, and upper arm areas as well as mild-moderate muscle loss of temporalis, deltoid, pectoralis, and interosseous muscle areas. He was also found to have significant pitting edema in both ankles and generalized mild edema, which may have masked further evidence of subcutaneous fat and muscle loss. The patient's abdomen was mildly distended, which his daughter reported has been his baseline since his cardiac arrest. He had a reported stage III pressure injury to the coccyx that was not directly assessed. The patient's daughter reported that the patient had been experiencing intermittent diarrhea. A retroperitoneal ultrasound showed increased echotexture of the kidneys consistent with medical renal disease and was negative for dilatation/obstruction; ascites was also seen. Serum and urine tests done upon admission are shown in Table 5.

Serum Urea Nitrogen/Creatinine Ratio

The patient's ratio of serum urea nitrogen to creatinine was markedly elevated. Increased serum urea nitrogen/creatinine ratios may occur as a result of AKI, gastrointestinal bleeding, advanced age, hypercatabolic conditions, steroid use, and hematoma resorption.¹⁹⁻²¹ In AKI, elevation of this ratio suggests prerenal injury secondary to dehydration or hypoperfusion.²² However, serum creatinine is reflective of somatic protein stores, and levels may be decreased among those with malnutrition and depleted muscle mass.¹⁶ Muscle atrophy is common among critically ill patients with prolonged ICU stays; this occurs as a result of increased muscle proteolysis and insufficient protein provision and synthesis.²³ There are a number of factors that trigger muscle proteolysis and reduce protein synthesis in critically ill patients, including immobilization/muscle inactivity, inflammation, cellular stress, and suboptimal protein intake.²³ It was theorized that during this patient's prolonged ICU

stay, these mechanisms produced muscle atrophy and resulted in a reduced serum creatinine level that was not reflective of kidney function. In addition, serum urea nitrogen concentrations may be increased in those with advanced age and higher severity of illness (conditions that contribute to a higher protein catabolic rate [PCR]), both factors that applied to the patient case.²⁴

Vitamin K Deficiency

The patient's prothrombin time (PT) was elevated at 18.3 seconds (see Table 5). The elevation was suspected to be secondary to vitamin K deficiency from a combination of malnutrition and antibiotic use. Vitamin K deficiency is a leading cause of PT prolongation in hospitalized patients.²⁵ Research in healthy adults suggests that subclinical vitamin K deficiency can develop within 2 weeks of ceasing consumption; this timeframe may be significantly reduced in critically ill patients, particularly those who have other risk factors.^{25,26} Most enteral formulas contain moderate amounts of vitamin K, which helps to prevent vitamin K deficiency if adequate enteral nutrition is delivered. However, if enteral feedings are frequently held, delayed, or infused at suboptimal levels, vitamin K provision may be inadequate to meet needs.

Antibiotic use has long been known to increase the risk of hemorrhagic disease because of the effect of antibiotics on vitamin K status.^{27,28} This occurs via 2 proposed mechanisms. Broad-spectrum antibiotics alter the microbial population within the gastrointestinal tract, effectively suppressing the growth of intestinal microbes that are necessary for the synthesis of vitamin K.²⁹ In addition, some antibiotics, including cefoperazone, moxalactam, cefamandole, and cefotetan, contain 1-N-methyl-5-thiotetrazole (NMTT); NMTT inhibits the carboxylation of the γ carbon of glutamic acid, which is a necessary step in the production of several clotting factors.³⁰ The patient was not reportedly prescribed any of the NMTT-containing antibiotics; however, he was prescribed multiple other antibiotics that may have affected vitamin K status through alteration of the intestinal microbiome. Additionally, he appears to have developed malnutrition as part of a prolonged hospitalization and critical illness.

Enteral Feeding Regimen

Determining the most appropriate enteral feeding regimen for adults with AKI can be complex. Patients with AKI often have multiple other comorbidities that can alter nutrition needs for various macronutrients and micronutrients. Malnutrition is a predictor of mortality among critically ill adults with AKI and must be ameliorated or prevented to help improve clinical outcomes.¹⁶ The patient's nutrition intake at the outside facility was unclear, based on available records and discussion with the patient's family; however,

Table 5. Blood and Urine Tests During Hospitalization in an 88-year-old male with AKI.

Test	Reference Range	Admission	Midpoint of Hospitalization	Final Lab Draw
Serum				
Serum urea nitrogen (mg/dL)	6–20	141	176	214
Creatinine (mg/dL)	0.6–1.2	1.7	2.4	3.4
Serum urea nitrogen/creatinine ratio	10–20.1	83	73	63
Potassium (mmol/L)	3.5–5.1	3.7	3.8	4.4
Phosphorus (mg/dL)	2.7–4.5	6.4	—	—
Calcium (mg/dL)	8.4–10.5	10.6	9.5	8.8
Prothrombin time (seconds)	10.0–13.5	18.3	18.4	—
Urine				
RBC (per HPF)	0–3	>100	—	—
Granular casts	Negative	Positive	—	—
Tubular cells	Negative	Positive	—	—
Urine sodium (mmol/24 h)	150–300	<20	—	—

AKI, acute kidney injury; HPF, high-powered field; RBC, red blood cell.

his bed-scale weight at the time of the current admission was consistent with his normal (pre-cardiac arrest) weight, despite his significant edema. This suggests that the patient's fluid retention may have been masking weight loss, which is considered a characteristic of malnutrition by the Academy of Nutrition and Dietetics (Academy) and American Society for Parenteral and Enteral Nutrition (ASPEN).³¹ The Academy/ASPEN malnutrition framework is used uniformly to identify malnutrition within the health system where this patient was hospitalized and was thus used in the assessment of this patient to determine the presence of malnutrition. In addition to significant edema, the patient also exhibited evidence of subcutaneous fat and muscle loss. Thus, based on the Academy/ASPEN malnutrition framework, the patient was considered to have moderate malnutrition in the context of chronic disease, as evidenced by mild-moderate subcutaneous fat and muscle loss, and fluid retention masking suspected weight loss. Identifying the presence and severity of malnutrition in hospitalized patients can help guide nutrition interventions; in this case, the presence of malnutrition, as well as a stage III pressure injury, warranted concerted effort to ensure adequate protein and calorie provision.

The patient's energy and protein needs were estimated to be approximately 1800 kcal (using Mifflin-St. Jeor equation with activity factor of 1.2) and approximately 100 g protein (1.2 g/kg actual body weight), respectively. This protein regimen was chosen as a conservative estimate to promote wound healing and restoration of nutrition status while not accelerating a need for RRT. Furthermore, because the patient was edematous, the team suspected that this protein dose would be an acceptable amount per kilogram of his dry weight. The patient was started on enteral feedings of a renal specialty formula with a goal of 45 mL/h for 22

hours daily, a protein modular (containing 100 kcal and 15 g protein) once daily, and autoflush of 25 mL water every 2 hours. Enteral feedings were held for 1 hour before and after levothyroxine administration, at the request of the patient's daughter. The prescribed regimen provided a total of approximately 1880 kcal, 95 g protein, ≈1000 mL free water, and 100% recommended daily intake of major vitamins and minerals daily. Though the protein modular was provided initially and recommended by the dietitian for wound healing and because of the underlying malnutrition, the patient's family declined the additional protein on the basis of the patient's AKI.

The patient tolerated the enteral feedings without the protein modular throughout his hospitalization; however, his kidney function continued to worsen, and the patient's mental status did not improve. He suffered worsening respiratory function and underwent multiple therapeutic thoracenteses. He also required a rapid response code due to mucous plugging. The family was presented with the decision of continuing with current medical care (which would have included the initiation of hemodialysis) or considering comfort-focused care with hospice. Ultimately, because of the patient's poor prognosis for meaningful recovery and continued deteriorating clinical condition, the family elected to change the patient's status to comfort-focused care, and the patient passed in the care of inpatient hospice.

This patient's hospitalization presented a number of nutrition issues, including the effect of his suspected malnutrition on serum creatinine (effectively masking the severity of his AKI); a suspected vitamin K deficiency affecting clotting ability; the question of how much protein to provide based on his kidney function, malnutrition, and significant wounds; and the end-of-life dilemmas affecting

nutrition-related decision making. Nutrition care in this case was provided in collaboration with the physician team and the patient's family.

Literature Review

A major concern when prescribing nutrition regimens for patients with AKI is preventing azotemia and further kidney damage that could theoretically occur with excessive protein provision. However, it has also been theorized that increased protein intake may improve GFR by enhancing renal blood flow.³² The balance of providing protein in an amount that is enough to improve renal perfusion and nitrogen balance but not contribute to azotemia and exacerbate kidney insult can be a challenge for nutrition support clinicians. In addition, a common nutrition obstacle in the ICU is the fact that what is actually provided is often significantly less than what is prescribed.^{33,34}

Search Strategy

A literature review was performed to search and assess the current data available regarding optimal protein provisions for patients with AKI who are not on RRT. The search was completed in June 2018 according to the search strategy shown in Table 6. Two databases (Pubmed and CINAHL) were searched for papers published within the past 15 years, using variations of the search terms “protein” and “acute kidney injury.” Primary studies were preferred, and review papers were excluded. With this search strategy, 3 resulting papers met inclusion criteria and were reviewed here. Hand searches of relevant journals (including *Journal of Renal Nutrition*, *Journal of Parenteral and Enteral Nutrition*, and *Nutrition in Clinical Practice*) were performed, and 1 additional paper meeting inclusion criteria was identified and included for review.

Summary of the Literature

The 4 studies analyzed in this review are described in Table 7. Sabatino et al published a recent observational study aimed at validating current predictive methods for estimating energy and protein needs among adults with AKI, as well as comparing actual energy and protein provisions with what was prescribed.³⁵ In this prospective study, energy and protein needs were estimated for adults with stage 3 AKI ($n = 43$) admitted to the ICU. Energy needs were estimated according to European Society for Parenteral and Enteral Nutrition (ESPEN) guidelines (25 kcal/kg), as well as Faisy-Faigon, Penn State, and Harris-Benedict equations. Protein needs were estimated according to ESPEN guidelines (1 g/kg for non-RRT AKI). In addition, energy and protein needs were measured using indirect calorimetry and

PCR, respectively. Actual intakes were then recorded for all patients. Results of the study showed a mean normalized PCR (nPCR) of 1.3 g/kg/d (or a mean total daily need of 107.9 ± 41.1 g) protein for adults with AKI not on RRT. Using ESPEN guidelines, protein needs were estimated at a mean of 87.1 ± 25.7 g daily; mean prescribed protein intake was 73.2 ± 40.0 g daily, and mean received protein intake was 66.2 ± 37.0 g daily. The measured protein need (using PCR) was significantly higher than what was prescribed or delivered ($P < .0001$). The authors concluded that protein delivery was inadequate for adults with AKI and that current predictive estimates for protein (and energy) were inaccurate in this patient population.³⁵

Kyle et al published a retrospective cohort study in 2017 to determine the effect of protein intake on renal recovery in children with AKI, as well as to compare protein and energy intakes before and after the implementation of a new nutrition guideline program.³⁶ Children ($n = 511$) admitted to the pediatric ICU (PICU) >48 hours were included; those with AKI had a mean age of 1.2 years. Energy needs were estimated using the Schofield equation, and protein needs were estimated using ASPEN pediatric guidelines (2–3 g/kg for ages 0–2 years). Energy and protein intakes from all sources (enteral nutrition, parenteral nutrition, and other intravenous sources) were calculated. Results from this study showed that median energy and protein intakes were less than the estimated requirements in all patient groups. The cumulative protein gap was significantly greater in those with persistent AKI compared with no AKI (median -1.1 g/kg/d, $P = .04$). Additionally, when actual protein intake met >80% estimated requirements, there was no increase in the risk for persistent AKI. The authors concluded that higher protein intakes were not associated with delays in renal recovery among pediatric patients with AKI.³⁶

A randomized controlled trial published in 2015 by Doig et al sought to evaluate the effect of an amino acid infusion on kidney function in critically ill adults.³⁷ Adult patients ($n = 466$) with an anticipated ICU stay of at least 2 days were included in the study. Participants were randomized to receive either “standard care” (according to standard hospital practices; specific macronutrient provisions not specified) or high-dose amino acids (2 g/kg IBW to a maximum of 100 g/d). A slightly higher proportion of the amino acid group had kidney dysfunction at baseline, though this difference was not statistically significant ($P = .07$). There was no significant difference between the 2 groups in the duration of renal dysfunction ($P = .78$), net fluid balance ($P = .35$), or need for RRT ($P = .21$). However, participants in the amino acid group did have significantly higher urine output ($P = .0002$) and higher GFR ($P = .004$) than those receiving standard care. The authors suggested that the increased amino acid provision may have promoted a diuretic effect (increasing urine output to accommodate increased

Table 6. Search Strategy for Literature Review.

Date	June 2018		
Inclusion criteria	• Subjects: Human • Age: Any • Setting: Hospital, ICU • Nutrition-related problem: AKI, not on RRT • Sample size: Any • Study dropout rate: <20% • Year range: 2003–2018 • Study preference: Primary studies • Authorship: If an author is included on >1 research article that is similar in content, the most recent article will be accepted and earlier versions will be rejected. If an author is included on >1 research article and the outcome is different, then both articles may be included.		
Exclusion criteria	• Language: English • Age: <18 years old • Setting: Ambulatory, outpatient • Study dropout rate: >20% • Year: Prior to 2003 • Language: Non-English • Study type: Review papers		
Database	Pubmed	CINAHL	Hand Search (relevant journals, reference lists)
Search Terms	“Acute kidney injury” (MeSH term) + “protein” (title/abstract) + “nutr*” (title/abstract)	“Protein” (title) + “renal failure, acute” (subject heading)	N/A
Hits	110	28	N/A
Total studies to review	3	0 (excluding duplicates)	1

AKI, acute kidney injury; ICU, intensive care unit; MeSH, medical subject headings; N/A, not applicable; RRT, renal replacement therapy. Note: Adapted from the Academy of Nutrition and Dietetics Evidence Analysis Manual.⁵³

fluid administration) that is generally not seen in critically patients.³⁸ They also concluded that the increased protein provision did not negatively affect kidney dysfunction or renal recovery, although a subgroup analysis in those with baseline kidney dysfunction was not performed.³⁷

In 2007, Singer et al published a small randomized controlled trial with the aim of testing the effect of a high-dose amino acid infusion on nitrogen balance and solute diuresis.³⁹ In this study, critically ill adults with nonoliguric AKI ($n = 14$) were randomized to receive either 75 g/d intravenous amino acids or 150 g/d intravenous amino acids for 3 days, in addition to 2000 kcal/d from dextrose and fats in both groups. Serum urea nitrogen increased significantly from baseline in the moderate-dose amino acid group but not in the high-dose group ($P < .04$). Those in the high-dose amino acid group required significantly less diuretic to achieve goal urine output ($P < .001$) and had a significantly greater cumulative nitrogen balance ($P < .001$). One patient (16%) in the moderate-dose amino acid group ultimately required RRT, whereas 2 patients (25%) in the high-dose amino acid group ultimately required RRT. No significant change in GFR was seen in either group. The authors concluded that high-dose amino acids appeared to enhance

diuretic effect and improve nitrogen balance but did not significantly affect renal function.³⁹

Discussion

There is a notable paucity of primary studies investigating the clinical question of how much protein should be provided to critically ill patients with AKI who are not on RRT. Protein needs in those requiring RRT have been more extensively studied, and guidelines for these patients are subsequently more specific.^{40–42} However, 95% of patients with AKI do not require RRT,^{43,44} and thus the identification of best practices for nutrition care in these patients warrants further attention.

Overall, the studies reviewed in this analysis suggest that increased protein provisions in critically ill patients with AKI who are not on RRT may be beneficial to kidney function and clinical outcomes. However, there is significant heterogeneity among the reviewed studies in the amount and route of protein provision, method of classifying AKI, participant ages, and outcomes measured. These differences among the studies weaken the overall conclusions that can be drawn from this literature review.

Table 7. Summative Table of Studies Investigating Protein Needs and Delivery in Adults With AKI Not on RRT.^{35-37,39}

Author and Year, Study Design, Class ^a	Study Purpose	Study Population	Interventions	Results	Conclusions	Strengths/Limitations
Sabatino 2017, Prospective observational, Class B	- To validate conventional predictive equations for energy and protein in patients with AKI, with or without RRT - To compare prescribed and received energy and protein with measured and estimated needs	- Adults (n = 43) with stage 3 AKI on nutrition support (with or without mechanical ventilation), admitted to ICU - Mean age 67 ± 15 years 72% male - Mean BMI 29 ± 9.4 kg/m²	- Daily nutrition prescriptions & intakes recorded until transferred out of ICU or transitioned to oral diet - Energy needs estimated according to ESPEN guidelines (25 kcal/kg), and Faisy-Faigon, Penn State, and Harris Benedict equations - Protein needs estimated according to ESPEN guidelines (1 g/kg for non-RRT, 1.7 g/kg for RRT) - Actual needs measured using IC (energy) and PCR (protein) q48 hr for the first week, then once weekly	- Among non-RRT patients: - Mean PCR (g/24 hr) was 107.9 ± 41.1 - Mean nPCR (g/kg) was 1.3 (range 0.6–2.7) - Mean calculated daily protein need was 87.1 ± 25.7 g - Mean prescribed daily protein intake was 73.2 ± 40.0 g - Mean received daily protein was 66.2 ± 37.0 g	- PCR values were significantly higher than both the prescribed and delivered protein amounts ($P < .0001$) - Protein intake was less than PCR > 75% of the time among non-RRT patients - Current methods of estimated protein (and energy) intake among patients with AKI are not accurate	Strengths: - High number of measurements - Multicenter, prospective design Limitations: - Relatively small sample size - Use of actual BW to estimate nutrition needs (potential for inaccuracy due to fluid retention)

(continued)

Table 7. (continued)

Author and Year, Study Design, Class ^a	Study Purpose	Study Population	Interventions	Results	Conclusions	Strengths/Limitations
Kyle 2017, Retrospective cohort, Class B	- To determine the effects of protein intake on AKI recovery - To compare protein and energy intakes in patients without and without AKI, before and after implementation of nutrition support guidelines	Children (n = 511, mean age 1.5 years without AKI and 1.2 years with AKI) admitted >48 hr to PICU	- Data collected retrospectively - Patients classified as AKI (using RIFLE criteria) or non-AKI - Enteral, parenteral, and IV intakes calculated - Nutrition Support Guidelines implemented using a 45-minute training session for all clinical PICU practitioners	30% (n = 156) of patients had AKI - Median energy & protein intakes were less than recommended in all patient groups - Protein intake in those with persistent AKI met 51% of estimated needs - The cumulative protein gap was significantly greater in those with persistent AKI compared with no AKI (median -1.1 g/kg/d, $P = .04$) - Protein intake of >80% estimated needs was not associated with increased risk of persistent AKI	- Higher protein intakes were not associated with a delay in renal recovery. - Protein intake was improved after implementation of nutrition support guidelines, but underfeeding persisted in those with AKI	Strengths: - Large sample size Limitations: - Retrospective data collection - Single center - Data on fluid balance and urine output not available - Use of predictive equations rather than IC

(continued)

Table 7. (continued)

Author and Year, Study Design, Class ^a	Study Purpose	Study Population	Interventions	Results	Conclusions	Strengths/Limitations
Doig 2015, RCT, Class A	To determine the effect of amino acid therapy on kidney function in critically ill adults	Adults (n = 466, mean age 62.7 years in control group and 63.3 years in amino acid group) in ICU	Participants were randomized to receive either 2 g/kg IBW of total protein (including IV amino acids + any other sources) to a maximum of 100 g/d IV amino acids, or standard care	- No significant difference between groups in duration of renal dysfunction ($P = .78$), net fluid balance ($P = .35$), or need for RRT ($P = .21$) - Amino acid group had significantly higher urine output ($P = .0002$) and higher GFR ($P = .004$)	IV amino acid administration did not affect duration of renal dysfunction but did improve GFR and urine output.	Strengths: - Study design (RCT) - Multicenter Limitations: - Unblinded
Singer 2007, RCT, Class A	To determine the effect of high-dose amino acids on nitrogen balance and solute diuresis	- Critically ill adults (n = 14) with non-oliguric AKI, requiring PN - Mean age 69.4 years	- Provided 2000 kcal/d nonprotein calories (dextrose + fat) - Group 1 received additional 75 g/d amino acids × 3 days - Group 2 received additional 150 g/d amino acids × 3 days - Nitrogen balance, biochemical tests, anthropometric measurements, and diuretic prescriptions were followed × 4 days; outcomes noted	- Serum urea nitrogen increased significantly in Group 1 ($P < .04$) but not in Group 2 - Amount of diuretic needed to achieve >3 L daily urine output was significantly less in Group 2 ($P < .001$) - Cumulative nitrogen balance significant greater in Group 2 ($P < .001$) - One patient (16%) in Group 1 ultimately required RRT; 2 patients (25%) in Group 2 ultimately required RRT	High-dose amino acids may improve nitrogen balance and reduce diuretic requirements, without increasing serum urea nitrogen, among nonoliguric patients with AKI	Strengths: - Study design (RCT) Limitations: - Duration of therapy - Mean age significantly different between groups

AKI, acute kidney injury; BMI, body mass index; ESPEN, European Society for Parenteral and Enteral Nutrition; GFR, glomerular filtration rate; hr, hour; IBW, ideal body weight; IC, indirect calorimetry; ICU, intensive care unit; IV, intravenous; nPCR, normalized protein catabolic rate; NSTI, necrotizing soft tissue injury; PCR, protein catabolic rate; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial; RIFLE, Risk, Injury, Failure, Loss of kidney function, End stage renal disease; RRT, renal replacement therapy. ^aAccording to the Academy of Nutrition and Dietetics Evidence Analysis Manual.⁵³

Though all of the reviewed studies examined critically ill patients, 3 studied adult patients,^{35,37,39} and 1 studied pediatric patients.³⁶ The pediatric study was still included in the analysis because of its strong study design and contribution to the overall discussion of protein needs in AKI. Three studies^{35,36,39} focused on patients with AKI at baseline, and 1 study³⁷ included a mix of participants with AKI and normal renal function at baseline. The definition of AKI also varied between these studies; this was not unexpected, as the system for defining and classifying AKI has recently evolved. Sabatino et al studied adults with stage 3 AKI according to the 2012 KDIGO guidelines.³⁵ Pediatric patients in the study by Kyle et al were identified as having AKI using the RIFLE criteria (risk, injury, failure, loss of kidney function, and end-stage kidney disease).³⁶ In the study by Singer et al, AKI was defined by a 50% decrease in GFR, a 100% increase in serum creatinine level, or an increase of serum creatinine to 3.5 mg/dL.³⁹ These participant differences may have contributed to differences in study results.

Methodological differences also existed between the reviewed studies. In the study by Sabatino et al, PCR was measured to determine protein needs among study participants; this was compared with estimated protein needs (using a 1 g/kg calculation according to ESPEN guidelines) and actual protein provision.³⁵ In the pediatric study by Kyle et al, protein needs were estimated using ASPEN guidelines (2–3 g/kg for children 0–2 years old; 1.5–2 g/kg for children 2–13 years old; 1.5 g/kg for children 13–18 years old).³⁶ Protein demands in pediatric patients would be expectedly higher than those for adults to account for anabolic processes associated with growth. Doig et al provided amino acid infusions in the dose of 100 g/d or a maximum of 2 g/kg daily; whether the amino acid dose was based on actual or IBW was not specified.³⁷ Singer et al provided amino acid infusions in either 75-g or 150-g daily doses.³⁹

The method used by Sabatino et al, in which PCR was measured to determine protein needs, most closely estimates and individualizes protein requirements.³⁵ This study found a mean nPCR of 1.3 g/kg (based on actual body weight) among adults with AKI who were not on RRT.³⁵ Current available recommendations for protein dosing in adults with AKI who are not on RRT are shown in Table 8. The most recently published guidelines from A.S.P.E.N and the Society of Critical Care Medicine suggest that standard recommendations for protein (1.2–2 g/kg based on usual body weight for normal weight patients or IBW for obese and critically ill patients) should be followed for adults with AKI who are not on RRT and that protein dose should not be restricted in these patients for the purpose of avoiding RRT.⁴² However, this specific guideline was based on expert consensus, with minimal supportive evidence. In a 2013 review paper, Fiaccadori et al suggest a protein target of

Table 8. Current Recommendations for Protein Dose in Adults With Acute Kidney Injury Not on RRT.^{11,42,46}

Source	Recommendation
SCCM/ASPEN Guidelines for the Provision and Assessment of Nutrition Support Therapy in the Adult Critically Ill Patient (2016)	• Standard ICU recommendations: 1.2–2 g/kg usual body weight (if normal weight) or IBW (if obese or critically ill) • Protein should not be restricted to delay or avoid RRT
KDIGO Clinical Practice Guideline for Acute Kidney Injury (2012)	• 0.8–1.0 g/kg • Protein should not be restricted to delay or avoid RRT
ESPEN Guidelines on Parenteral Nutrition: Adult Renal Failure (2009)	• 0.6–1.0 g/kg
ESPEN Guidelines on Enteral Nutrition: Adult Renal Failure (2006)	• 0.6–1.0 g/kg⁵⁴

ASPEN, American Society for Parenteral and Enteral Nutrition; ESPEN, European Society of Parenteral and Enteral Nutrition; IBW, ideal body weight; ICU, intensive care unit; KDIGO, Kidney Disease Improving Global Outcomes; RRT, renal replacement therapy; SCCM, Society of Critical Care Medicine.

0.8–1 g/kg (based on usual body weight for normal weight patients or IBW for overweight patients) in noncatabolic adults with AKI who are not on RRT.⁴⁵ This recommendation also appears to be consensus-driven, without cited supportive evidence. Other consensus-based guidelines also suggest 0.8–1.0 g/kg/d in noncatabolic adults with AKI not requiring RRT, but again with limited experimental basis.^{11,46}

Measured outcomes also varied between the studies analyzed in this review. Sabatino et al compared the measured and estimated protein requirements with the protein dose that was prescribed and actually provided; clinical outcomes were not assessed in this study.³⁵ In the pediatric study by Kyle et al, study investigators similarly compared estimated protein needs with actual protein provisions but also looked at the associated risk of persistent AKI at different levels of protein intake.³⁶ Doig et al studied the duration of renal dysfunction (using serum creatinine levels), estimated GFR, serum cystatin C, net fluid balance, urine output, and need for RRT as measures of effects on kidney function.³⁷ Doig et al also looked at a number of other clinical outcomes, including performance status, physician function, length of ICU stay, length of total hospitalization, mechanical ventilation days, and organ dysfunctions.³⁷ Singer et al measured serum urea nitrogen levels, creatinine levels, sodium

excretion, nitrogen balance, diuretic requirements, and need for RRT among study participants receiving different amino acid doses.³⁹

Among the studies that evaluated clinical outcomes associated with protein provisions, results suggested overall that increased protein provision is not associated with adverse health effects. Kyle et al found that among pediatric patients with AKI, higher protein intakes (ie, 80% of normal recommendations for critically ill children: at least 1.6–2.4 g/kg for children ages 0–2 years, 1.2–1.6 g/kg for children ages 2–13 years, and 1.2 g/kg for children ages 13–18 years old) were not associated with delays in renal recovery.³⁶ Note that in this study, protein intakes were dichotomized into “higher” (ie, 80% of estimated requirement) and “lower” (<80% of estimated requirement); thus, a “higher” protein intake was not one that was increased above the normal recommendations for critically ill children but rather was a description of how much protein was actually provided vs prescribed. In other words, what these results suggest is that providing less protein was not renal-protective.

Using high-dose intravenous amino acids compared with standard nutrition treatment, Doig et al found that increased protein provision did not change the duration of AKI but did improve GFR and urine output.³⁷ High-dose amino acids were associated with higher fluid intake but also increased urine output, which suggests that the amino acid infusion may have promoted a diuretic effect. Fluid overload is a major concern for critically ill patients, particularly for those with impaired renal function; in critical illness, increased fluid administration is not generally met with a compensatory increase in urine output.³⁸ Thus, strategies that facilitate the body’s natural diuretic response to increased fluid administration can help to prevent fluid overload.³⁷ Singer et al also found clinical benefits from intravenous amino acids infusions; high-dose amino acids (150 g/d) were associated with improved nitrogen balance and reduced diuretic requirements, without increases in serum urea nitrogen, among critically ill adults with non-oliguric AKI.³⁹ More patients in the high-dose amino acid group (2 patients in the high-dose group compared with 1 patient in the moderate-dose group) eventually required RRT, though the statistical significance of this difference was not noted. This was a small study and may have been under-powered to adequately analyze this variable.

Based on the above analysis, it is clear that further research is needed to determine the optimal protein dose for patients with AKI who are not on RRT. This question has clinical significance, as there are risks to providing too little as well as too much protein in this patient population. Negative nitrogen balance has been associated with increased mortality risk among critically ill patients with AKI.⁴⁷ Additionally, studies have shown that actual nutrition support provision (particularly with enteral feedings) is often significantly less than what is prescribed by a

clinician.^{33,34} For this reason, and the increased catabolism that is associated with critical illness and its myriad of comorbid conditions, the risk of malnutrition in the ICU is high.⁴⁸ Prescribing a restricted protein dose for a critically ill patient with AKI, if there is no indication to do so, has the potential for negative nutrition and clinical consequences. However, there is valid concern that providing excessive protein doses to patients with impaired renal function may increase urea generation and contribute to azotemia; whether this would result in the need for RRT or impair long-term renal recovery is unclear. Another consideration is that protein may actually aid renal function by improving renal perfusion. An early study published by Abel et al in the *New England Journal of Medicine* found that, in critically ill adults with AKI (n = 53), short-term intravenous infusion of essential amino acids actually led to faster renal recovery.⁴⁹ Finally, it must be noted that AKI rarely exists as an isolated condition; it often occurs in conjunction with other comorbid conditions associated with critical illness, including lung disease, cardiovascular conditions, gastrointestinal impairments, wounds, infections, etc. The consideration of the patient’s entire condition, including kidney function, other comorbid conditions, hemodynamic stability, and so forth, must be part of the nutrition assessment and management process. Future research should focus on the optimal estimation and provision of protein and its effect on immediate kidney function, clinical outcomes, and long-term renal recovery.

Conclusion

In this patient case, malnutrition and muscle loss resulted in reduced creatinine generation and a falsely low serum creatinine level. This resulted in an elevated ratio of serum urea nitrogen to creatinine and masked the severity of the patient’s AKI. The effect of muscle mass on serum creatinine levels can be profound and is a consideration that the nutrition support clinician must make when evaluating nutrition needs.

There are many factors that impact protein needs in critically ill adults with AKI. Research investigating the optimal dose of protein for patients with AKI but not on RRT is limited. Current recommendations guiding clinical practice are largely consensus-based, and experimental data testing the effect of protein on renal function and clinical outcomes in this population are needed. Malnutrition is a significant risk for critically ill patients and can cause a host of adverse clinical outcomes.^{50,51} Unnecessary restriction of protein (or overall nutrition support) in this population may pose risks to the patient; however, excessive protein provision in this population could theoretically have a negative effect on renal function. Future studies should be aimed at identifying the optimal protein dose for adults with AKI who are not on RRT to help improve nutrition

and clinical outcomes without impairing kidney function or recovery.

Statement of Authorship

E. Schwartz, R. Hillyer, and J. Ziegler contributed to the conception and design of the research; E. Schwartz, R. Hillyer, and J. Foley contributed to the acquisition and analysis of the data; E. Schwartz, J. Ziegler, and K. Willcutts contributed to the interpretation of the data; E. Schwartz drafted the manuscript. All authors critically revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

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